

Indic Diarization and ASR: Benchmark and Improvement

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Summary. The best benchmarked combination is **IndicConformer-600M** × **pyannote 3.1** (DER 27.34, JER 38.14, cpWER 79.67, WDER 20.12). The adopted improvement, a **language-ID fallback** to Whisper, lowers cpWER to **72.96** (−6.71) and WER to **71.71** (−7.11): 12 clips better, 87 unchanged, **0 worse**, and it holds on all three diarizers. DER/JER are unchanged by construction, since the diarizer never sees words. **LLM speaker relabelling** was built and measured, and rejected: it worsens cpWER and WDER (§3.2).

1 Data and metrics

- **Audio.** 99 of 100 segments extracted (one video is no longer available): 12.26 h, 16 kHz mono, trims verified sample-exact. 9 scripts, 2–8 speakers, 7.61% of speech overlapped.
- **References.** 38 turns outside the window dropped, 88 clipped, 1 inverted and 3 empty turns removed; 666 bracket tags (<noise>, <laughter>, ...) stripped before scoring.
- **Metrics.** DER/JER with pyannote.metrics at **collar 0, overlap scored**, UEM = full clip; cpWER with MeetEval [12, 14]; JER as in DIHARD II [11]; WDER hand-rolled after El Shafey et al. [2]. Rates are error-weighted; a 0.25 s collar with overlap excluded would report pyannote at 20.58, not 27.34.
- **Ground truth never enters the pipeline;** the one reference-diarization condition is a labelled diagnostic.

2 Baselines

Diarizer	DER	JER	Miss	FA	Conf.	Spk-count acc.
pyannote 3.1 [1, 8]	27.34	38.14	11.59	5.89	9.86	72.7%
Sortformer 4spk-v1, streaming [6]	47.23	57.82	9.82	6.28	31.13	51.5%
Sortformer 4spk-v1, offline*	74.85	65.29	65.14	2.19	7.52	43.4%

*Its $O(T^2)$ attention ran out of memory on the 25 longest clips on a T4; those score as total miss, so this row measures memory, not the model.

- **Speaker count.** pyannote is flat from 2 to 5+ speakers (DER 25.9–29.7); streaming Sortformer rises from 34.4 to 57.4 under its hard 4-speaker cap, and detects speech better (miss 9.82) but assigns it $3\times$ worse (confusion 31.13).
- **Overlap is not where most error lies.** pyannote’s DER is 46.47 inside overlapped speech and 20.41 elsewhere, but only **24%** of its error seconds fall in overlap.

ASR design: transcribe the whole clip, then assign words. Each clip is transcribed once; every word goes to the diarizer turn it overlaps most, and words with no turn are kept (dropping them would reward a diarizer for missing speech). Against ASR on each diarized segment, this makes every cpWER difference purely a labelling difference, avoids Whisper’s weakness on sub-second fragments, and needs 99 ASR calls, not 12,809. A word inside overlap goes to the speaker covering more of it and is counted separately; given the 24% error share, source separation was not built first.

ASR (× pyannote 3.1)	WER	cpWER	WDER
IndicConformer-600M multilingual (AI4Bharat [5]; Conformer [4], CTC, ONNX)	78.82	79.67	20.12
Whisper large-v3 [10] (faster-whisper, greedy)	85.70	86.67	21.27

Two decoding defects were fixed before benchmarking.

1. **IndicConformer’s CTC head is multi-softmax:** one softmax per language block, so scores across blocks are not comparable. A global argmax spelled words across several scripts. Choosing the clip’s language by frame vote and decoding within that block takes **WER from 93.66 to 78.82**; the naive decode is kept as a scored ablation.
2. **faster-whisper’s temperature fallback samples without a seed:** one clip returned 87, 84 and 100 words on three runs, and 211 every time at temperature 0. Corpus WER improved 86.62→85.70 and runtime fell from 396 to 153 min.

cpWER sits less than one point above WER: **recognition, not attribution, is the bottleneck.**

3 Improving the output

3.1 Language-ID fallback (adopted)

Diagnosis. IndicConformer decodes each clip in one language. On 13 clips its own vote chose **Urdu (11) or Nepali (2)**, so every word came out in the wrong script and the clip scored 100% WER regardless of what was heard. Hindi and Urdu speech is highly similar and conventionally written in different scripts, so this is acoustic ambiguity rather than a decoder bug. **Rule.** If IndicConformer’s detected language is outside the served set {hi, mr, bn, gu, kn, ml, or, pa, ta, te}, use Whisper’s words for that clip. No reference is read.

IndicConformer × ...	pyannote 3.1	Sortformer streaming	Sortformer (74 clips)
WER	78.82 → 71.71	78.82 → 71.71	75.41 → 67.69
cpWER	79.67 → 72.96	86.96 → 82.99	81.65 → 76.89
WDER	20.12 → 9.32	39.58 → 34.62	24.12 → 19.46
cpWER per clip: better / same / worse	12 / 87 / 0	12 / 86 / 1	8 / 66 / 0

- **Devanagari WER falls from 79.2 to 57.2**, better than either system alone (Whisper: 69.4). Every fallback row reproduces exactly the per-clip scores of the system it took its words from.

- **Not fitted to scores:** routing Hindi- or Marathi-detected clips to Whisper instead makes corpus WER *worse* (82.56); the gain is the out-of-set rule, not a language preference.
- **Caveats:** the served-language list is task knowledge; part of the WDER drop reflects WDER being ill-conditioned near 100% WER.

3.2 Speaker relabelling from the transcript (not adopted)

Design, after DiarizationLM [13] and lexical speaker error correction [7]. Transcripts are cut into units (same-speaker runs split at pauses > 0.5 s); a unit may only move to an existing speaker, so **text cannot change and WER must not move** (asserted per clip). The LLM, Qwen2.5-7B-Instruct [9] in 4-bit on a T4, sees windows of 25 units and returns JSON edits with a confidence; edits below 0.7 are dropped and a clip where it edits over 30% of units is discarded. A **rule baseline** moves any unit under 1 s to its nearer neighbour.

IndicConformer $\times$ pyannote 3.1	cpWER	WDER	WDER per clip: better / same / worse
baseline	79.67	20.12	—
+ rule	79.70	20.01	30 / 31 / 38
+ LLM	80.00	20.96	6 / 74 / 19

Why it failed.

1. **Ceiling:** even after pause splitting, **37.87% of words** sit in units spanning two true speakers, out of reach of any relabel.
2. **Weak evidence:** at $\sim 79\%$ WER the text says little about who is speaking. The LLM’s confidences were always 0.8 or 0.9, and it did most damage on 2-speaker clips (WDER 7.54 $\rightarrow$ 12.16).
3. **ASR-dependent effect:** the rule helps all 6 IndicConformer conditions and hurts all 3 Whisper ones. CTC output fragments into more, shorter units (4,468 vs. 3,132 on the same turns), where a sub-second unit is usually diarizer jitter.

DER for a word-level fix. Mapping word labels back onto the diarizer’s turns with *zero* edits already raises DER from 27.34 to 29.03, so that projection is the fair control. Against it the rule adds +1.52 DER and the LLM +0.87; the results table reports the control beside every DER.

4 Indic-specific observations

- **Script, not sound, decides the score:** Hindi/Urdu and Gujarati/Nepali language-ID confusions turn a plausible transcript into 100% WER.
- **The two systems fail in opposite places.** IndicConformer: Kannada 64.9, Odia 68.5, Malayalam 71.4 WER, where Whisper is at 96–101 (over 100 through insertions); Whisper wins on Devanagari (69.4 vs. 79.2) and Tamil (78.1 vs. 92.1). IndicConformer mostly deletes: 0.42 output words per reference word, often dropping characters inside otherwise correct words.
- **Reference-turn attribution scores worse WDER than pyannote** (21.45 vs. 20.12): word timings drift past the tight reference turns, suggesting timing mismatch in the labels.

5 Limitations and next steps

- **Limitations.** Reference labels were not hand-audited (the timing result above is the strongest hint they need it); the served-language list is a deployment assumption; video was unused, though active-speaker detection targets exactly the dominant single-speaker confusion.
- **Next.** Restrict IndicConformer’s language vote to the served set; combine both ASR systems at word level (ROVER [3]); separate speakers in overlap; rerun relabelling on the improved ASR.

Reproducibility. `notebooks/end_to_end.ipynb` runs every stage in order on Colab: it rebuilds the references from the spreadsheet byte-for-byte (SHA-256), re-runs each model live on a 3-clip subset, and recomputes attribution, scoring and the results table over all 99 clips, asserting they equal the committed tables. The subset reproduces the reference run exactly on other hardware: **0.00% DER for all three diarizers, identical words for all three ASR systems**. Full 99-clip GPU runs are one switch away but need > 5 T4h, so they ran on Kaggle (notebooks with outputs in `notebooks/`); extraction ran from a home connection, as YouTube bot-gates cloud IPs. Per-video tables: `data/results/results_table.md`, `results_per_video.xlsx`.

References

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